Assignment 3

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Section 1: Introduction

In this report, we analyze a data set of the amount of total text messages a person typed for participants at two different time points for participants categorized into two groups. The data is found in the file called *TextMessages.csv*. First, we load and prepare the data for analysis. Then we create 2 visualizations and compute summary statistics. The purpose of this report is to describe the *TextMessages.csv* data.

Section 2: Prepare and Review Data for Analysis

```
#Load data in TextMSG object
#Do not need to set working directory because working in project

TextMSG <- read.csv("TextMessages.csv")

#_____________________________________________________________________________
#Importing libraries to be used throughout the data

#install.packages("tidyverse")
library (tidyverse)
#install.packages("ggplot2")
library(ggplot2)

#_____________________________________________________________________________
#Looking at the data and understanding how it is broken down in the file

is.data.frame (TextMSG)
```

```
## [1] TRUE
```

```
#The response of "True" suggests that our data is a data frame

str(TextMSG)
```

```
## 'data.frame':    50 obs. of  4 variables:
##  $ Group      : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ Baseline   : int  52 68 85 47 73 57 63 50 66 60 ...
##  $ Six_months : int  32 48 62 16 63 53 59 58 59 57 ...
##  $ Participant: int  1 2 3 4 5 6 7 8 9 10 ...
```

```
#This indicates structure and we can see that we have 50 objects with 4
#different variables and data types. Based on this information,
#we have a total of 50 observations, four columns, and all the
#variables (baseline, group, Six_month, and participant) are stored as integers.
```

```r
#In addition, we can see that our dataset is in the wide format with each
#participant having separate columns for measurements taken at different times.
#In addition it shows that each participant has two measurements: baseline and
#Six_months

#Checking if it is a factor:
is.factor (TextMSG$Group)
```

```
## [1] FALSE
```

```r
#This confirms that our group variable needs to be adjusted to be a categorical
#variable. To double check what the groups look like:
table(TextMSG$Group)
```

```
##
##  1  2
## 25 25
```

```r
#This shows us that we have 2 groups total, each with 25 observations.
#We can now convert Group into a factor:

TextMSG$Group <- factor (TextMSG$Group)
#Double checking that it now converted:
is.factor(TextMSG$Group)
```

```
## [1] TRUE
```

```r
head(TextMSG)
```

```
##   Group Baseline Six_months Participant
## 1     1       52         32           1
## 2     1       68         48           2
## 3     1       85         62           3
## 4     1       47         16           4
## 5     1       73         63           5
## 6     1       57         53           6
```

```r
#based on how the first 6 lines of the data look, we can also see that the
#"Participant" variable is just the random number assigned to a participant.
#This gives a clearer breakdown of what the data looks like.

#Since our data is in the wide format, we may do better by changing this into
#long format to help with analysing this data

#To convert the data into the long format we can do the following:
#We will use tidyverse, which was already imported above

TextMSG_long <- TextMSG %>% pivot_longer (cols = c(Baseline, Six_months),
                                          names_to = "Time",
                                          values_to = "TextMessages")

#View (TextMSG_long)
#This confirms that we now have 100 observations, which means that our data was
#transformed to the longer format with a new variable called Time containing the
#original column names (combining Baseline and Six_months into a single column)

#We also can check what the data now looks like:
```

```r
head (TextMSG_long)
```

```
## # A tibble: 6 x 4
##   Group Participant Time       TextMessages
##   <fct>       <int> <chr>             <int>
## 1 1               1 Baseline             52
## 2 1               1 Six_months           32
## 3 1               2 Baseline             68
## 4 1               2 Six_months           48
## 5 1               3 Baseline             85
## 6 1               3 Six_months           62
```

```r
#We see that the data has been transformed.

#Since the variable Time is also two measurement, we want to check if that is a
#factor as well, and we look at the levels to identify if it needs to be adjusted:
is.factor (TextMSG_long$Time)
```

```
## [1] FALSE
```

```r
#This confirms that it is not, so we can convert it:

TextMSG_long$Time <- factor (TextMSG_long$Time,
                             levels = c("Baseline", "Six_months"))

is.factor(TextMSG_long$Time)
```

```
## [1] TRUE
```

```r
#Now we see that the variable Time has been converted into a factor.

#Double checking that all of the variables are in the correct formats:
str(TextMSG_long)
```

```
## tibble [100 x 4] (S3: tbl_df/tbl/data.frame)
##  $ Group       : Factor w/ 2 levels "1","2": 1 1 1 1 1 1 1 1 1 1 ...
##  $ Participant : int [1:100] 1 1 2 2 3 3 4 4 5 5 ...
##  $ Time        : Factor w/ 2 levels "Baseline","Six_months": 1 2 1 2 1 2 1 2 1 2 ...
##  $ TextMessages: int [1:100] 52 32 68 48 85 62 47 16 73 63 ...
```

```r
#We can see that we have two variables that are factors (group and time) and
#two variables that are still integers (participant and TextMessages)
```

Section 3: Analysis

In this section, we create two visualizations and compute summary statistics. First, we create a faceted boxplot to show the distribution of text messages by group and time. Next, we create a faceted bar plot to visualize the mean number of text messages by group and time. Finally, we review the summary statistics.

Section 3.1: Visualization One

```r
#Visualization 1
#The first visualization is focused on looking at boxplots by group and time.

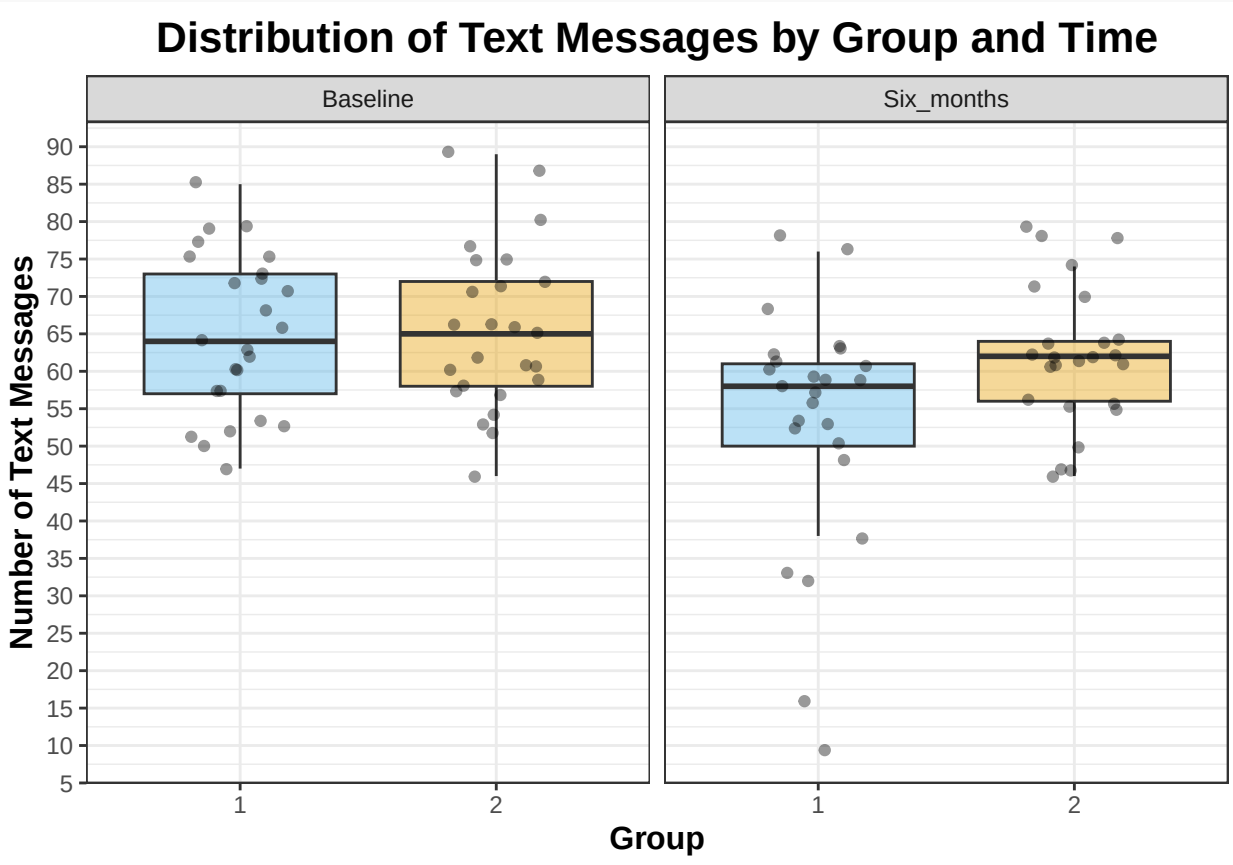
#Faceted Boxplot
```

```r
#using ggplot2, which was loaded earlier
#This plot also has the data points plotted in the background to make it clearer
#to show the spread.

boxplot_TextMSG_long <- ggplot(TextMSG_long, aes(x=Group,
                                                 y = TextMessages,
                                                 fill = Group))+
  geom_boxplot(alpha =0.4, outlier.shape = NA)+
  geom_jitter(width = 0.20, alpha = 0.4) +
  facet_wrap(~Time) +
  scale_fill_manual(values = c("#56B4E9", "#E69F00")) +
  labs (title = "Distribution of Text Messages by Group and Time",
        x = "Group",
        y = "Number of Text Messages" ) +
  theme_bw() +
  theme(plot.title = element_text(hjust = 0.5, size = 16, face = "bold"),
        axis.title = element_text(hjust = 0.5, size = 12, face = "bold"),
        plot.subtitle = element_text(hjust = 0.5, size = 12, face = "bold"),
        legend.position = "none") +
  scale_y_continuous(breaks = seq(from =0, to = 95, by=5))

#Since we are plotting the points, we took out the outliers so they are not
#plotted twice

#View the plot:
boxplot_TextMSG_long
```



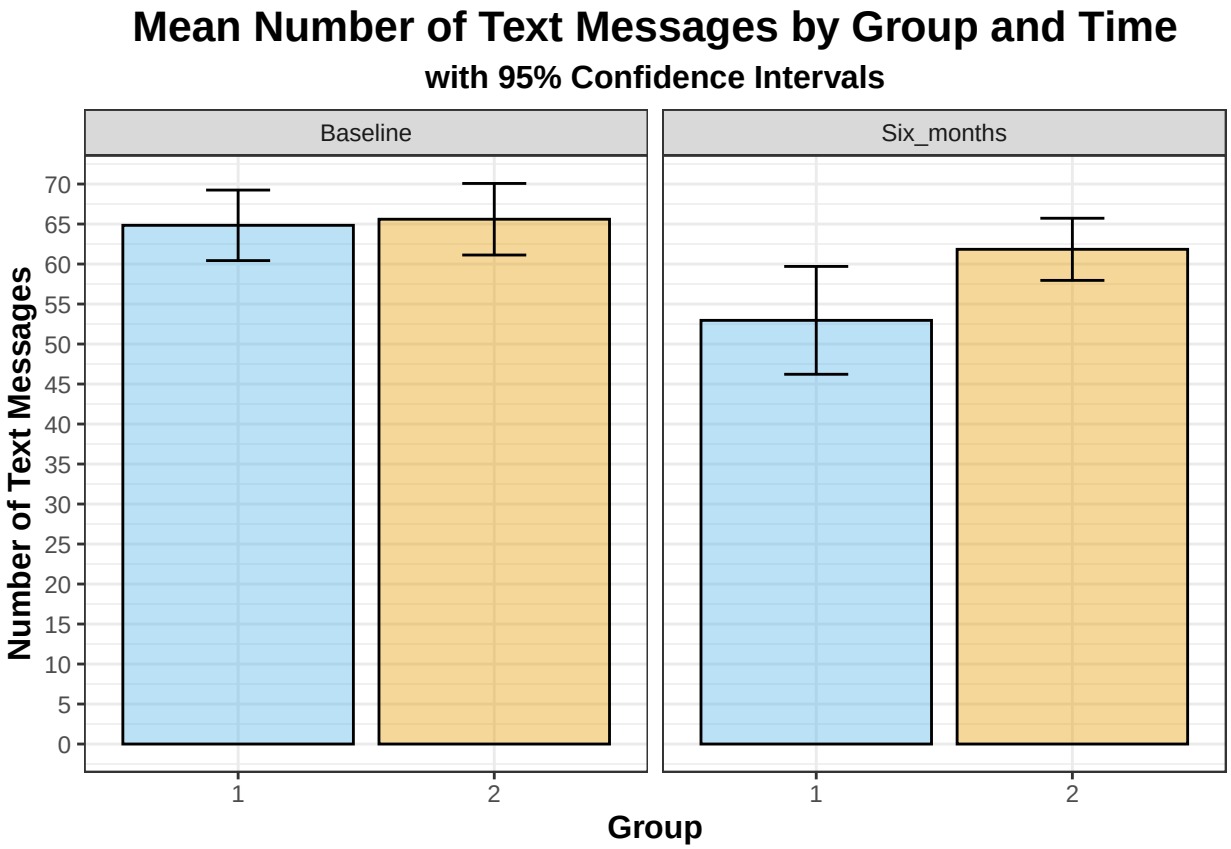
Based on the box plot, at the baseline, both groups have very similar medians and also look like they have similar spread and distribution of text messages, with no visible outliers. Their median numbers of messages were approximately 64 for Group 1 and 65 for Group 2. In addition, their boxes overlapped considerably, indicating similar central tendencies and variability. However after 6 months, there is a visual difference between the two groups. At six months, the median decreased in both groups. Group 1 decreased to approximately 58 messages, whereas Group 2 decreased to approximately 62 messages. Group 1 also showed greater variability at six months and had several unusually low observations, including values below 20 messages. The outliers for group 1 appear to be more spread out and both above and below group data. In comparison, Group 2 had a somewhat higher median and a more tightly clustered distribution at six months. Overall, the graph suggests a larger decrease and greater variability for Group 1, although a statistical test is needed as these are just visual observations.

Section 3.2: Visualization Two

```
#Visualization 2

#Create barplot with desired formatting, using color blind accessible colors,
#and assign to barplot_TextMSG_long object:
barplot_TextMSG_long <- ggplot(TextMSG_long,
                               aes(Group, TextMessages, fill = Group)) +
  stat_summary(fun = mean, geom = "bar", colour = "Black", alpha = 0.4) +
  stat_summary(fun.data = mean_cl_normal, geom ="errorbar", colour = "black",
               width = 0.25) +
  facet_wrap(~Time) +
  scale_fill_manual(values = c("#56B4E9", "#E69F00")) +
  labs(x="Group", y="Number of Text Messages",
       title = "Mean Number of Text Messages by Group and Time",
       subtitle = "with 95% Confidence Intervals") +
  theme_bw() +
  theme(plot.title = element_text(hjust = 0.5, size = 16, face = "bold"),
        axis.title = element_text(hjust = 0.5, size = 12, face = "bold"),
        plot.subtitle = element_text(hjust = 0.5, size = 12, face = "bold"),
        legend.position = "none") +
  scale_y_continuous(breaks = seq(from =0, to =70, by=5))

#Display bar plot:
barplot_TextMSG_long
```



The bar plots depict the mean number of text messages by group and time with superimposed error bars in black. The error bars represent the 95% confidence intervals about the mean.

The bar plots show that the mean number of text messages for the groups at the baseline time are both around 65. The plots at six months show Group 1 had a noticeable drop in number of text messages with a mean around 53 while the mean for Group 2 dropped to about 62. The confidence intervals at six months overlap partially, so the plot alone does not establish whether the groups differ significantly; a formal statistical test would be needed to draw that conclusion.

Section 3.3: Summary Statistics

```
#Summary statistics

#Summarize text messages by group and time
SummaryTable_byGroupTime <- TextMSG_long %>%
  group_by(Group, Time) %>%
  summarise(n=n(),
            min = min(TextMessages), max = max(TextMessages),
            range = max(TextMessages) - min(TextMessages),
            mean=mean(TextMessages),
            SD=sd(TextMessages),
            median=median(TextMessages),
            Q1=quantile(TextMessages, 0.25),
            Q3=quantile(TextMessages, 0.75),
            .groups = "drop")

knitr::kable(SummaryTable_byGroupTime, format = "markdown",
```

```
            digits = 2,
            caption = "Summary Statistics of Text Messages by Group Time")
```

Table 1: Summary Statistics of Text Messages by Group Time

Group	Time	n	min	max	range	mean	SD	median	Q1	Q3
1	Baseline	25	47	85	38	64.84	10.68	64	57	73
1	Six_months	25	9	78	69	52.96	16.33	58	50	61
2	Baseline	25	46	89	43	65.60	10.84	65	58	72
2	Six_months	25	46	79	33	61.84	9.41	62	56	64

Note: the SD describes the spread of individual text message counts, while the error bars in Visualization 2 show 95% confidence intervals for each mean.

The summary statistics describe the number of text messages by group and time.

At baseline, the two groups are very similar. Group 1 has a mean of 64.8 (SD = 10.7) and a median of 64, and Group 2 has a mean of 65.6 (SD = 10.8) and a median of 65.

At six months, both groups sent fewer text messages, but the drop was larger for Group 1. Group 1's mean fell to 53.0 and its median fell to 58, while Group 2's mean fell to 61.8 and its median to 62.

At six months, Group 1's mean (53.0) is noticeably lower than its median (58), and its standard deviation increased to 16.3, which is consistent with the low outliers shown in the boxplot in Visualization 1. These outliers pull the mean down for Group 1 at six months. Group 1's middle 50% of values (Q1 to Q3: 50 to 61) became more concentrated at six months, even as SD increased. Group 2's spread decreased slightly at six months (SD = 9.41, Q1 to Q3: 56 to 64).

These means are consistent with the bar chart in Visualization 2.

Section 4: Conclusion

This assignment focused on creating 2 different visualizations using ggplot() for both box plot and bar plot. Both plots show a decrease in text messages from baseline to six months.

Based on the summary statistics and the plots, at baseline, the two groups were very similar in their number of text messages, with means of 64.8 for Group 1 and 65.6 for Group 2. At six months, the mean decreased in both groups, but the reduction was larger in Group 1. Group 1 decreased by 11.8 messages on average, from 64.8 to 53.0, whereas Group 2 decreased by 3.8 messages, from 65.6 to 61.8. Group 1 also showed greater variability at six months, with its standard deviation increasing from 10.7 to 16.3. In contrast, Group 2 remained more consistent, with its standard deviation decreasing slightly from 10.8 to 9.41. Overall, these descriptive results suggest that Group 1 experienced a larger reduction in text messages and more variation among participants than Group 2. However, an inferential statistical test would be required to determine whether these changes and the difference between groups are statistically significant.